

Quiz — 1.2.3 Software Development

Name: _____ Date: _ **Mark:** ____ / 25

OCR H446 · Component 01 · 1.2.3

Section A — Multiple choice (8 marks)

1. Which methodology is **linear/sequential**, completing one stage fully before the next begins? A. Agile B. Waterfall C. Spiral D. RAD Your answer: ☐
 2. A defining feature of the **Agile** approach is: A. A single delivery of the finished product at the end B. Iterative increments with continuous client feedback C. Heavy upfront documentation with no later changes D. Risk analysis at the centre of every loop Your answer: ☐
 3. **Pair programming** and test-driven development are most associated with: A. Waterfall B. Extreme programming (XP) C. Spiral D. RAD Your answer: ☐
 4. The **Spiral** model is best suited to projects that are: A. Small with fixed, clear requirements B. Large, expensive and high-risk C. Needed extremely quickly with vague requirements D. Trivial throwaway scripts Your answer: ☐
 5. **RAD** primarily relies on: A. Detailed sequential documentation B. Rapid prototyping refined by user feedback C. A central risk-analysis loop D. One fixed requirements specification Your answer: ☐
 6. During which lifecycle stage are **requirements gathered** and **feasibility** assessed? A. Design B. Analysis C. Implementation D. Evaluation Your answer: ☐
 7. Testing a program with **boundary data** means using values that are: A. Far outside the acceptable range B. At the edges of the acceptable range C. Always invalid/erroneous D. Random and unrepeatable Your answer: ☐
 8. The **Evaluation** stage judges the finished system mainly against: A. The personal preferences of the lead programmer B. The original requirements (effective, usable, robust, maintainable) C. The number of lines of code written D. The cost of the hardware Your answer: ☐
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Section B — Short answer (12 marks)

9. State **two** advantages and **one** disadvantage of the Waterfall methodology. [3]

10. Explain why Agile is often more suitable than Waterfall when a client's requirements are **unclear or likely to change**. [2]

11. Name and describe the three categories of test data used to test a program thoroughly. [3]

12. Describe **two** tasks carried out during the **Design** stage of the software lifecycle. [2]

13. Explain what is meant by **risk analysis** in the Spiral model and why it benefits a high-cost project. [2]

Section C — Exam-style question (5 marks)

14. A startup is building a brand-new mobile app. The client is unsure exactly what features they want and expects ideas to change as early versions are shown to real users. The budget is modest and the team is small but skilled. Recommend a **suitable development methodology** and justify your choice with reference to the scenario. [5]

Answer key

Section A (8 marks — 1 each) 1. **B** — Waterfall is linear/sequential, one stage at a time. 2. **B** — Agile delivers iterative increments with continuous client feedback. 3. **B** — XP adds pair programming and constant/test-driven testing to Agile. 4. **B** — Spiral suits large, expensive, high-risk projects via repeated risk analysis. 5. **B** — RAD centres on rapid prototyping refined by user feedback. 6. **B** — Analysis investigates the problem, gathers requirements and checks feasibility. 7. **B** — boundary data lies at the edges of the acceptable range. 8. **B** — Evaluation compares the system against the original requirements.

Section B (12 marks)

9. [3] Advantages (1 each, max 2): simple/easy to manage; well documented; clear structure with defined milestones. Disadvantage (1): inflexible / hard to go back and change; the client only sees the finished product at the **end**, so errors in requirements are costly.
10. [2] Agile works in iterative increments and gathers continuous client feedback (1), so changing or newly discovered requirements can be built into the next iteration rather than being fixed at the start as in Waterfall (1).
11. [3] **Normal/valid** data — typical values within range that should be accepted (1); **boundary** data — values at the edges of the acceptable range (1); **erroneous/invalid** data — values outside the range that should be rejected (1).
12. [2] Any two (1 each): design data structures; design algorithms; design the user interface / screen layouts; design inputs, outputs and file/data formats.
13. [2] Risk analysis = at each loop the team identifies and evaluates possible risks and plans how to reduce them before continuing (1); on a high-cost project this catches expensive problems early, reducing the chance of costly failure later (1).

Section C (5 marks) 14. [5] Recommendation must be justified against the scenario clues. Best answers name **Agile** (or RAD). Up to 5 from: requirements are **unclear/likely to change**, so an iterative methodology that adapts each cycle is suitable (1); **continuous client/user feedback** lets the evolving ideas be incorporated as early versions are shown to real users (1); rapid prototyping/increments deliver working versions quickly for review (1 — especially if RAD named); the **small, skilled team** can cope with the lighter documentation Agile requires (1); fits the **modest budget**/short timescale better than a heavily documented Waterfall, which would lock requirements too early (1). (Reasoned justification with scenario links required for full marks; pure description caps at ~2.)

Total: 8 + 12 + 5 = 25